

Customer Shopping Behavior Analysis

Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.

Task is to analyze the company's consumer behavior dataset to answer the following overarching business question:
"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN
Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases											
3900	3900.000000	3900	3900											
2	NaN	6	7											
No	NaN	PayPal	Every 3 Months											
2223	NaN	677	584											
NaN	25.351538	NaN	NaN											
NaN	14.447125	NaN	NaN											
NaN	1.000000	NaN	NaN											
NaN	13.000000	NaN	NaN											
NaN	25.000000	NaN	NaN											
NaN	38.000000	NaN	NaN											
NaN	50.000000	NaN	NaN											

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Download:** Downloaded and data loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MS SQL server to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
1	Male	157890
2	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
1	3	64
2	5	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	average_review_rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Skirt	3.78

5. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping

	average_purchase	shipping_type
1	60.48	Express
2	58.46	Standard

6. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	average_revenue	total_revenue
1	Yes	1053	59.49	62645
2	No	2847	59.87	170436

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
1	Hat	50
2	Coat	49
3	Sneakers	49
4	Sweater	48
5	Pants	47

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history

50 % No issues found

	customer_segment	Number of customers
1	Loyal	3116
2	Returning	701
3	New	83

7. **Top 3 Products per Category** – Listed the most purchased products within each category.

	category	item_ranks	item_purchased	total_orders
1	Accessories	1	Jewelry	171
2	Accessories	2	Sunglasses	161
3	Accessories	3	Belt	161
4	Clothing	1	Blouse	171
5	Clothing	2	Pants	171
6	Clothing	3	Shirt	169
7	Footwear	1	Sandals	160
8	Footwear	2	Shoes	150
9	Footwear	3	Sneakers	145
10	Outerwear	1	Jacket	163
11	Outerwear	2	Coat	161

8. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

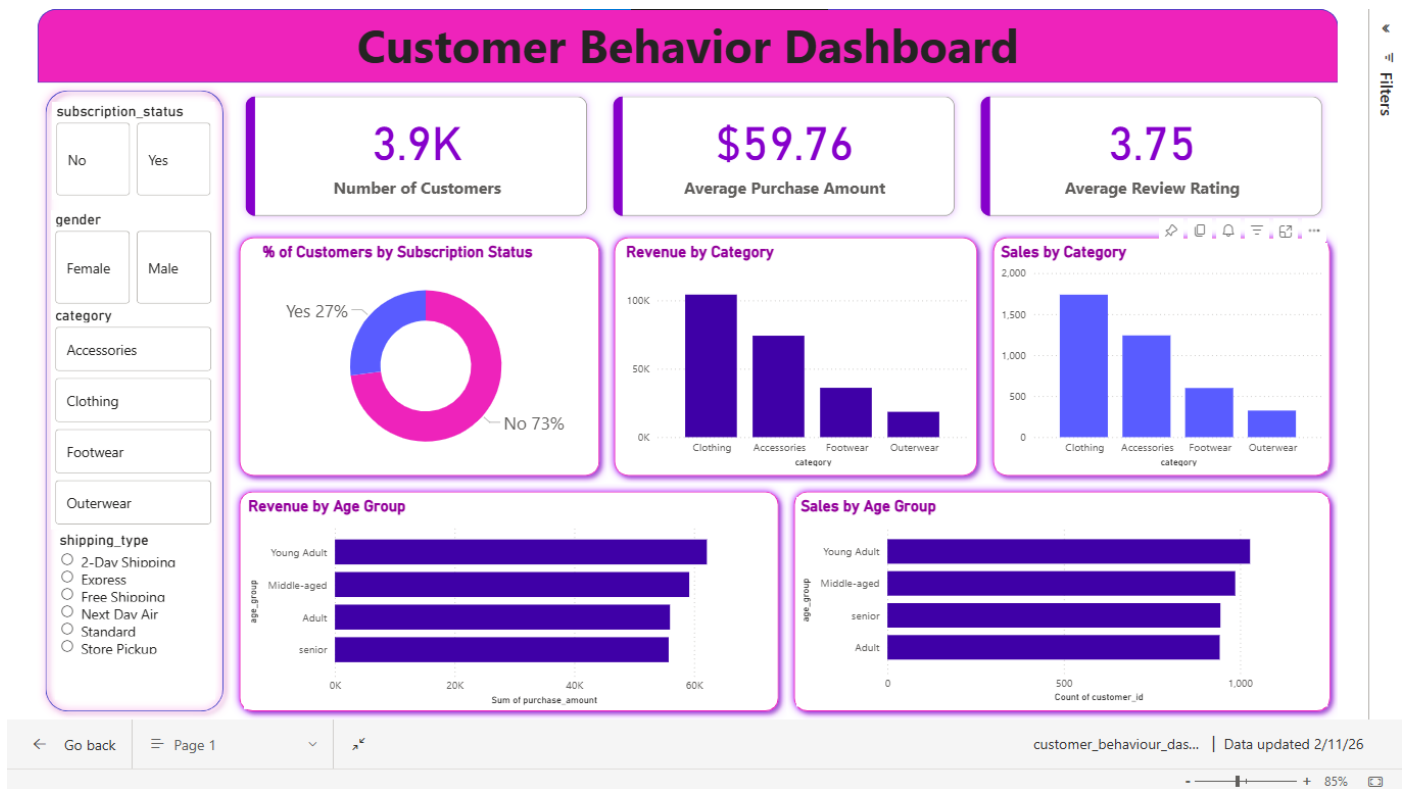
	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group

	age_group	revenue_contribution
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.